

```

1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPi.py
2 The network model is cPIKAN
3 Layers: ModuleList(
4   (0): ChebyKANLayer(in_features=1, out_features=10, degree=1)
5   (1): ChebyKANLayer(in_features=10, out_features=10, degree=1)
6   (2): ChebyKANLayer(in_features=10, out_features=1, degree=1)
7 )
8 Total trainable parameters: 240
9 Model on device: cuda:0
10 Start training...
11 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
12 Training: 100%|██████████| 2000/2000 [00:08<00:00, 226.10epoch/s, Loss=8.
  21e-03, LR=1.0e-03, Best=8.21e-03]
13 Training finished. Best loss: 3.44e-03. Model saved to results/best_model.pth
14 Loss history saved to results/loss_history.npy
15 Training time: 9.687410593032837s
16 Relative loss: 0.0752912387251854
17 L2 loss: 0.04790791869163513
18 Parameter containing:
19 tensor([[-1.0467,  1.6929],
20         [ 1.0005, -1.4346],
21         [-1.0178,  1.1259],
22         [-0.0319,  0.8312],
23         [-1.0465,  1.5011],
24         [ 0.8192, -1.2560],
25         [-0.5443,  1.6875],
26         [-0.1007,  2.3854],
27         [-0.8435, -0.4561],
28         [-0.4052,  2.0500]]], device='cuda:0', requires_grad=True)
29 Parameter containing:
30 tensor([[-3.9913e-02, -2.4216e-01],
31         [ 6.2369e-02, -4.0390e-01],
32         [ 4.3633e-02, -1.6945e-01],
33         [-1.9024e-02,  3.1349e-01],
34         [-8.0106e-02, -1.9056e-01],
35         [-2.7840e-02,  4.9242e-01],
36         [ 1.2036e-01, -4.6910e-01],
37         [-5.6168e-02, -1.6065e-01],
38         [ 1.2471e-01, -9.4536e-01],
39         [-2.4281e-02, -6.4110e-01]],
40
41         [[-1.0184e-02,  8.0459e-02],
42          [-3.2643e-02,  8.5948e-02],
43          [-2.3649e-03,  1.5213e-01],
44          [ 7.8513e-02, -8.4160e-02],
45          [-1.7214e-02, -4.8095e-03],
46          [ 1.6028e-02, -2.1782e-01],
47          [-4.1353e-02,  2.3217e-01],
48          [ 7.6579e-02,  3.5516e-03],
49          [ 9.6720e-02,  4.4295e-01],
50          [ 6.8210e-02,  1.5120e-01]],
51

```

```

52      [[ 4.0420e-02,  1.4147e-01],
53      [ 2.3821e-02, -4.3981e-02],
54      [-2.3596e-02,  3.3011e-02],
55      [ 3.4592e-02, -1.3857e-03],
56      [-6.9926e-03, -3.5121e-02],
57      [ 2.3924e-03,  1.3947e-01],
58      [ 1.1887e-03, -1.1119e-01],
59      [-4.6584e-02, -1.0081e-01],
60      [ 1.1142e-01, -2.7192e-01],
61      [-3.3010e-02,  3.5908e-02]],
62
63      [[-8.3803e-02, -1.7635e-01],
64      [ 4.0691e-02, -7.7851e-02],
65      [ 2.4979e-02, -5.1294e-02],
66      [ 5.6214e-02,  1.1332e-01],
67      [ 1.2730e-02, -1.1160e-01],
68      [ 2.9199e-02,  2.0706e-01],
69      [-4.7583e-02, -1.2197e-01],
70      [-1.8525e-02, -6.4985e-03],
71      [ 1.0828e-01, -2.1192e-01],
72      [ 5.8448e-02, -1.2270e-01]],
73
74      [[ 4.9092e-02, -6.8597e-03],
75      [-9.6656e-03, -1.9097e-01],
76      [ 2.8625e-02, -8.8710e-02],
77      [ 7.6382e-02,  2.0801e-01],
78      [-4.9657e-02,  1.0493e-01],
79      [-7.3341e-02,  1.7772e-01],
80      [ 3.6913e-02, -2.4728e-01],
81      [-3.1054e-02, -6.0068e-03],
82      [ 9.5386e-02, -4.8572e-01],
83      [ 1.2485e-02, -1.7645e-01]],
84
85      [[-3.8828e-02,  2.7032e-03],
86      [-1.2000e-01,  1.4694e-01],
87      [-8.9529e-02,  8.0774e-02],
88      [ 6.1652e-02, -1.9922e-01],
89      [-5.6303e-02, -2.7938e-04],
90      [ 5.6496e-02, -1.8712e-01],
91      [ 1.3760e-03,  2.5158e-01],
92      [-1.2095e-02,  2.6782e-02],
93      [-1.5422e-02,  6.0404e-01],
94      [ 9.2182e-02,  1.0667e-01]],
95
96      [[-4.1803e-02, -4.7180e-01],
97      [ 1.9587e-02, -8.5918e-02],
98      [-1.2807e-02, -2.0398e-02],
99      [ 3.7959e-02,  1.6483e-02],
100     [-2.2297e-02, -4.1725e-01],
101     [ 5.8642e-02,  7.9849e-02],
102     [-4.5118e-02, -6.1815e-02],
103     [-9.2566e-02, -3.1267e-01],
104     [ 8.8679e-02, -3.9274e-01],

```

```

105      [ 8.1776e-02, -7.6053e-01]],
106
107      [[-9.0961e-02, -8.7063e-01],
108      [-1.1201e-01,  4.4772e-02],
109      [-5.0745e-02,  1.7813e-01],
110      [ 2.4025e-02, -1.5227e-01],
111      [ 1.0967e-01, -7.1182e-01],
112      [ 6.2596e-02, -2.1089e-02],
113      [ 2.3883e-02, -3.8828e-03],
114      [-2.9793e-02, -4.8200e-01],
115      [ 6.3690e-02, -1.5169e-01],
116      [ 1.7531e-02, -9.2356e-01]],
117
118      [[-7.2731e-04,  6.4121e-02],
119      [ 3.3778e-02,  1.0063e-01],
120      [-6.8821e-02,  6.9422e-02],
121      [-2.3975e-02, -1.1440e-01],
122      [ 1.4560e-03, -9.3970e-02],
123      [-1.2934e-01, -5.1395e-02],
124      [ 8.7828e-03,  2.8809e-02],
125      [ 5.5830e-02, -6.7227e-04],
126      [ 1.2537e-01, -5.2362e-02],
127      [ 6.6933e-02,  1.0716e-01]],
128
129      [[-3.5021e-02, -9.7590e-01],
130      [-5.5451e-02, -6.4388e-02],
131      [-5.7364e-02,  2.3135e-01],
132      [ 1.4531e-02,  5.1530e-02],
133      [-8.0421e-02, -8.0714e-01],
134      [-2.3740e-02,  6.5871e-02],
135      [-4.4472e-02, -6.9512e-02],
136      [-8.0295e-02, -7.2456e-01],
137      [ 1.4209e-01, -5.4245e-01],
138      [ 9.1570e-02, -1.2907e+00]]], device='cuda:0', requires_grad=True)
139 Parameter containing:
140 tensor([[[[-0.0297, -0.2511]],
141
142      [[-0.0662,  0.2466]],
143
144      [[ 0.0009,  0.1047]],
145
146      [[ 0.0039, -0.2688]],
147
148      [[-0.0588, -0.1937]],
149
150      [[-0.0258, -0.2309]],
151
152      [[ 0.0324,  0.2068]],
153
154      [[ 0.0382, -0.0338]],
155
156      [[-0.0177,  0.4538]],
157

```

```

158      [[-0.0462, -0.4287]]], device='cuda:0', requires_grad=True)
159 =====
160 Layer (type:depth-idx)          Output Shape          Param #
161 =====
162 CPIKANModel                    [1, 1]                --
163 |---ModuleList: 1-1            --                        --
164 |   |---0.weight                |---20
165 |   |---1.weight                |---200
166 |   |---2.weight                |---20
167 |   |---ChebyKANLayer: 2-1      [1, 10]               20
168 |   |   |---weight              |---20
169 |   |---ChebyKANLayer: 2-2      [1, 10]               200
170 |   |   |---weight              |---200
171 |   |---ChebyKANLayer: 2-3      [1, 1]                20
172 |   |   |---weight              |---20
173 =====
174 Total params: 240
175 Trainable params: 240
176 Non-trainable params: 0
177 Total mult-adds (Units.MEGABYTES): 0.00
178 =====
179 Input size (MB): 0.00
180 Forward/backward pass size (MB): 0.00
181 Params size (MB): 0.00
182 Estimated Total Size (MB): 0.00
183 =====
184
185 进程已结束, 退出代码为 0
186

```